Step-3 Weekly Software Development Progress Report

1. Implementation Progress

Following the Week 3 integrated RAG workflow, two areas were improved:

- **Answer traceability and retrieval (FR-08, FR-12):** Increased the number of retrieved passages from 3 to 5 in src/app.py. Updated src/generation/answer_generator.py to require PDF page references for factual statements, avoid invented page numbers, and distinguish PDF page numbers from academic citations.
- **PDF validation and error handling (FR-02, FR-13):** Updated src/ingestion/pdf_loader.py to detect password-protected, invalid/corrupted, and zero-page PDFs. Updated src/app.py to display clearer error messages.

These improvements extend the existing Week 3 workflow without changing the overall RAG architecture.

2. Integration and Testing

The updated PDF validation and answer-generation components were tested with the existing application.

- Four academic research papers were used to test answer generation, retrieval, and page references.
- Automated tests increased from 11 to 13 passing tests.
- Manual testing covered normal PDFs, blank/scanned PDFs, password-protected PDFs, invalid/corrupted PDFs, and zero-page PDFs.
- The application displayed the expected messages for unsupported or invalid PDF inputs.
- No failing automated tests remained after the changes.

Evidence: tests/test_pdf_loader.py, other test files under tests/, and implementation modules under src/. The test suite completed with 13 passing tests using uv run pytest.

3. GitHub Repository Changes

The following changes were committed and merged into main:

- feadc0f – Improved answer traceability and retrieval. Modified src/app.py and src/generation/answer_generator.py on branch answer-traceability.
- 29a75ad – Merged pull request #2.
- febdb39 – Improved PDF validation and error handling. Modified src/app.py, src/ingestion/pdf_loader.py, and tests/test_pdf_loader.py on branch pdf-validation.
- 3e14caf – Merged pull request #3.
- No additional documentation files under docs/step_2/ were changed during these commits. Earlier milestone documents were preserved.

4. Obstacles and Next Week's Priorities

The project still needs further evaluation of retrieval relevance and answer accuracy. Scanned PDFs are not supported, and handling unavailable Ollama models remains outstanding.

Priorities for next week:

1. Improve error handling when required Ollama models are unavailable.
2. Conduct a structured evaluation of retrieval relevance, answer accuracy, and page-reference correctness using academic PDFs and sample questions.
3. Perform additional end-to-end testing and document the results.

Expected deliverables: Improved model-availability error messages, documented evaluation results, and verified end-to-end application behavior.